Outline

1. TD1 – Python Foundations, BSM Greeks & Monte Carlo
2. TD2 – Gradient Descent & Logistic MLE
3. TD3 – Time Series: AR/ARIMA/GARCH & OLS
4. TD4 – Value at Risk, Backtesting & CoVaR
5. TD5 – Markowitz Portfolio Optimization
6. TD6 – Index Construction
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Log Returns: Definition

Let P_t be the price at time t. The **log return** (continuously compounded return) is

$$\boxed{r_t^{\log} = \ln\left(\frac{P_t}{P_{t-1}}\right)}$$

Using $\ln(a/b) = \ln(a) - \ln(b)$, this is equivalently

$$\boxed{r_t^{\log} = \ln(P_t) - \ln(P_{t-1})}$$

Relationship with Simple Returns

The simple return is $R_t = \dfrac{P_t - P_{t-1}}{P_{t-1}} = \dfrac{P_t}{P_{t-1}} - 1$, so $1 + R_t = \dfrac{P_t}{P_{t-1}}$.
Substituting into the log-return definition:

$$\boxed{r_t^{\log} = \ln(1 + R_t)}$$

Conversely, exponentiating both sides, $e^{r_t^{\log}} = 1 + R_t$, so

$$\boxed{R_t = e^{r_t^{\log}} - 1}$$

Time-Additivity of Log Returns

The cumulative log return between dates 0 and T is

$$\sum_{t=1}^{T} r_t^{\log} = \sum_{t=1}^{T} \ln\left(\frac{P_t}{P_{t-1}}\right) = \ln\left(\frac{P_1}{P_0}\right) + \ln\left(\frac{P_2}{P_1}\right) + \cdots + \ln\left(\frac{P_T}{P_{T-1}}\right)$$

Using $\ln(a) + \ln(b) = \ln(ab)$, all intermediate prices telescope/cancel:

$$\sum_{t=1}^{T} r_t^{\log} = \ln\left(\frac{P_1}{P_0}\frac{P_2}{P_1}\cdots\frac{P_T}{P_{T-1}}\right)$$

$$\boxed{\sum_{t=1}^{T} r_t^{\log} = \ln\left(\frac{P_T}{P_0}\right)}$$

So the cumulative log return is simply the **sum** of the individual log returns.

Approximation for Small Returns

Recall the first-order approximation $\ln(1+x) \approx x$ for x close to 0.
Since $r_t^{\log} = \ln(1+R_t)$, for sufficiently small simple returns:

$$\boxed{r_t^{\log} \approx R_t}$$

For small daily returns, simple and log returns are numerically close, though never exactly equal.

Aside: Weighting a Histogram

Without `weights`, a histogram bin's height is just its observation count n_j. Using weights $w_i = 1/N$ (with N the sample size), bin j's height becomes

$$\text{Height}_j = \sum_{i \in \text{bin } j} w_i = \sum_{i \in \text{bin } j} \frac{1}{N} = \boxed{\frac{n_j}{N}}$$

Since $\sum_j n_j = N$, this gives $\boxed{\sum_j \text{Height}_j = 1}$: each bin shows the **proportion** of observations it contains.

Contrast with `density=True`

$$w_i = 1/N: \sum_j \text{Height}_j = 1 \qquad \text{vs.} \qquad \texttt{density=True}: \sum_j \text{Height}_j \times \text{Width}_j = 1$$

The first normalizes bin heights to proportions; the second normalizes total *area* to a probability density.

The AR(1) Process and the Stationarity Condition

Consider $\boxed{X_t = c + \phi X_{t-1} + \varepsilon_t}$, with $\varepsilon_t \sim WN(0, \sigma_\varepsilon^2)$.
An AR model uses the **past** to describe and predict the **future**: for this to remain meaningful, the statistical properties of the process must stay stable over time. For the AR(1) process to be covariance-stationary, we require

$$\boxed{|\phi| < 1}$$

Why $|\phi| < 1$ Matters: the MA(∞) Representation

Substituting $X_{t-1} = c + \phi X_{t-2} + \varepsilon_{t-1}$ into $X_t = c + \phi X_{t-1} + \varepsilon_t$ gives

$$X_t = c(1+\phi) + \phi^2 X_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t$$

Repeating this substitution recursively,

$$X_t = c(1 + \phi + \phi^2 + \cdots) + \varepsilon_t + \phi\varepsilon_{t-1} + \phi^2\varepsilon_{t-2} + \cdots$$

If $|\phi| < 1$, then $\boxed{\phi^k \to 0}$ as $k \to \infty$ (distant shocks vanish), and the geometric series converges: $1 + \phi + \phi^2 + \cdots = \dfrac{1}{1-\phi}$. Hence the AR(1) admits the infinite moving-average representation

$$\boxed{X_t = \frac{c}{1-\phi} + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j}}$$

Constant Mean and Constant Variance

Taking expectations of the MA(∞) form (using $\mathbb{E}[\varepsilon_t] = 0$):

$$\boxed{\mathbb{E}[X_t] = \frac{c}{1-\phi}} \qquad \text{(does not depend on } t\text{)}$$

Since the innovations are uncorrelated, $\text{Var}(X_t) = \sigma_\varepsilon^2 \sum_{j=0}^{\infty} \phi^{2j}$. Because $|\phi| < 1$, $\sum_{j=0}^{\infty} \phi^{2j} = \dfrac{1}{1-\phi^2}$, so

$$\boxed{\text{Var}(X_t) = \frac{\sigma_\varepsilon^2}{1-\phi^2}} \qquad \text{(also does not depend on } t\text{)}$$

Autocovariance and the Definition of Stationarity

For a stationary AR(1), the autocovariance at lag h is

$$\boxed{\gamma(h) = \mathrm{Cov}(X_t, X_{t-h}) = \phi^{|h|} \frac{\sigma_\varepsilon^2}{1-\phi^2}}$$

– it depends only on the lag $|h|$, never on the calendar date t. These three properties define **covariance stationarity**:

$$\boxed{\mathbb{E}[X_t] = \mu, \qquad \mathrm{Var}(X_t) = \sigma_X^2, \qquad \mathrm{Cov}(X_t, X_{t-h}) = \gamma(h)}$$

Economic Interpretation: Shocks Die Out

A shock ε_t affects the process one period later by ϕ, two periods later by ϕ^2, and h periods later by ϕ^h:

$$\boxed{\text{Effect of a shock after } h \text{ periods} = \phi^h}$$

If $|\phi| < 1$, then $\phi^h \to 0$: shocks have only a **temporary** effect. This is **mean reversion** – the process drifts back toward its long-run mean $\boxed{\mu = \dfrac{c}{1-\phi}}$.

What Happens When $\phi = 1$: the Random Walk

Setting $\phi = 1$, $c = 0$: $X_t = X_{t-1} + \varepsilon_t$. By recursive substitution,

$$\boxed{X_t = X_0 + \sum_{j=1}^{t} \varepsilon_j}$$

Since the innovations are uncorrelated, $\mathrm{Var}(X_t) = \mathrm{Var}(X_0) + t\,\sigma_\varepsilon^2$:

$$\boxed{\mathrm{Var}(X_t) = \mathrm{Var}(X_0) + t\sigma_\varepsilon^2}$$

The variance now **grows with** t: the process is non-stationary. Moreover $\phi^h = 1^h = 1$ for all h, so

a shock has a permanent effect on a random walk

Stationary AR(1) versus Random Walk

	$\lvert\phi\rvert < 1$	$\phi = 1$
Process	Stationary AR(1)	Random walk
Effect of shock after h periods	$\phi^h \to 0$	1
Mean reversion	Yes	No
Variance	$\sigma_\varepsilon^2/(1-\phi^2)$	grows with t
Long-run distribution	Stable	Not stable

Stationarity ensures the statistical dynamics remain stable over time.

Extension to an AR(p) Process

For $X_t = c + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t$, write $\phi(L) X_t = c + \varepsilon_t$ with lag operator L and $\phi(L) = 1 - \phi_1 L - \cdots - \phi_p L^p$.
The process is stationary when every root z_j of the characteristic equation

$$\boxed{1 - \phi_1 z - \phi_2 z^2 - \cdots - \phi_p z^p = 0}$$

lies **outside the unit circle**: $\boxed{|z_j| > 1 \text{ for every root}}$. For AR(1), $1 - \phi z = 0 \Rightarrow z = 1/\phi$, and $|z| > 1 \iff |1/\phi| > 1 \iff \boxed{|\phi| < 1}$ – recovering the familiar condition.

From AR(1) to the Dickey–Fuller Regression

For $X_t = \phi X_{t-1} + \varepsilon_t$: stationary if $|\phi| < 1$; a **unit root** ($\phi = 1$) gives the random walk $X_t = X_{t-1} + \varepsilon_t$ (non-stationary).
Subtracting X_{t-1} from both sides: $X_t - X_{t-1} = (\phi - 1)X_{t-1} + \varepsilon_t$. Defining $\Delta X_t = X_t - X_{t-1}$ and $\gamma = \phi - 1$:

$$\boxed{\Delta X_t = \gamma X_{t-1} + \varepsilon_t}$$

Testing $\phi = 1$ is now equivalent to testing $\gamma = 0$.

The ADF Test: Hypotheses and Regression

$$
\begin{aligned}
H_0 &: \gamma = 0 \iff \phi = 1 \Rightarrow \text{unit root / non-stationary} \\
H_1 &: \gamma < 0 \iff \phi < 1 \Rightarrow \text{stationary}
\end{aligned}
$$

The **Augmented** DF test adds lagged differences (to whiten the error term), and optionally a trend:

$$
\Delta X_t = \alpha + \lambda t + \gamma X_{t-1} + \sum_{j=1}^{p} \delta_j \Delta X_{t-j} + \varepsilon_t
$$

Interpreting the ADF Test

H_0 : series has a unit root	H_1 : series is stationary (around included deterministics)

ADF result	Interpretation
p-value $< \alpha$	Reject $H_0 \Rightarrow$ evidence against a unit root
p-value $\geq \alpha$	Fail to reject $H_0 \Rightarrow$ insufficient evidence

Equivalently, comparing statistics directly: ADF stat $<$ critical value $\Rightarrow$ reject H_0 ("smaller" means *more negative*).

Careful

p-value ≥ 0.05 does **not** prove non-stationarity – it only means we lack sufficient evidence against the unit-root null.

ADF: Key Point

$$H_0 : \text{unit root / non-stationarity}$$
$$H_1 : \text{stationarity}$$
$$p\text{-value small} \implies \text{reject the unit-root hypothesis}$$

In finance, price **levels** often behave like a unit-root process, while **returns** or log-returns are much more commonly stationary.

Best Linear Prediction

Let $X_1, \ldots, X_n, Y$ be random variables. The **best linear predictor** of Y from $X_1, \ldots, X_n$ minimizes expected squared error:

$$\boxed{L(Y \mid X_1, \ldots, X_n) = \arg \min_{\hat{Y} = b_0 + b_1 X_1 + \cdots + b_n X_n} \mathbb{E}\big[(Y - \hat{Y})^2\big]}$$

i.e. choosing $b_0^\star, \ldots, b_n^\star = \arg\min_{b_0,\ldots,b_n} \mathbb{E}\big[(Y - b_0 - b_1 X_1 - \cdots - b_n X_n)^2\big]$, giving the prediction error (residual)

$$\boxed{Y - L(Y \mid X_1, \ldots, X_n)}$$

– the part of Y that cannot be linearly explained by $X_1, \ldots, X_n$.

Partial Correlation

To measure the relationship between Y and Z **after removing** the linear effect of $X_1, \ldots, X_n$: predict both, keep the residuals

$$\boxed{e_Y = Y - L(Y \mid X_1, \ldots, X_n)} \qquad \boxed{e_Z = Z - L(Z \mid X_1, \ldots, X_n)}$$

The **partial correlation** is the correlation between these residuals:

$$\boxed{\text{Partial correlation} = \text{Corr}\left(e_Y, e_Z\right)}$$

Intuition: instead of the total correlation between Y and Z, we first strip out whatever is explained by the control variables.

Why Ordinary Autocorrelation Is Not Enough

The ordinary autocorrelation $\rho_2 = \text{Corr}(X_t, X_{t-2})$ measures **total** association – but part of it may be transmitted indirectly:

$$\boxed{X_{t-2} \longrightarrow X_{t-1} \longrightarrow X_t}$$

So X_t and X_{t-2} can be correlated even if X_{t-2} adds nothing *once* X_{t-1} *is known*. At lag n, the PACF asks: *after controlling for* $X_{t-1}, \ldots, X_{t-n+1}$*, does* X_{t-n} *still carry extra linear information about* X_t*?*

Defining the PACF

For a stationary series, the best linear prediction of X_t from its last n lags is (coefficients independent of t):

$$\boxed{L(X_t \mid X_{t-1}, \ldots, X_{t-n}) = \phi_{n0} + \phi_{n1} X_{t-1} + \cdots + \phi_{nn} X_{t-n}}$$

The coefficient on the *last* lag, ϕ_{nn}, is the partial autocorrelation at lag n. The PACF is the sequence

$$\boxed{\text{PACF}(n) = \phi_{nn}, \qquad n = 1, 2, 3, \ldots}$$

Interpreting ϕ_{nn}

$$\boxed{\phi_{nn} = \text{partial correlation between } X_t \text{ and } X_{t-n}, \text{ controlling for } X_{t-1}, \ldots, X_{t-n+1}}$$

Correlation at lag n : total linear association between X_t and X_{t-n}
Partial corr. at lag n : direct, incremental association only

PACF Examples at Lags 1, 2, 3

Lag 1: no intermediate lags to remove, so $\boxed{\phi_{11} = \text{Corr}(X_t, X_{t-1}) = \rho_1}$.

Lag 2: remove the effect of X_{t-1}:

$$\boxed{\phi_{22} = \text{Corr}\left(X_t - L(X_t \mid X_{t-1}),\ X_{t-2} - L(X_{t-2} \mid X_{t-1})\right)}$$

Lag 3: similarly, remove the effects of both X_{t-1} and X_{t-2}. The PACF isolates the **incremental** information of each additional lag.

The Cut-Off Property for an AR(p) Process

If the true process is $\boxed{X_t = \phi_0 + \phi_1 X_{t-1} + \cdots + \phi_p X_{t-p} + \varepsilon_t,\ \phi_p \neq 0}$, then X_t depends directly only on the last p lags. Older lags may still be **correlated** with X_t (information propagates through the chain $X_{t-3} \to X_{t-2} \to X_{t-1} \to X_t$), but they add **no direct** linear contribution once the relevant lags are controlled for. Hence

$$\boxed{\phi_{kk} = 0, \qquad k > p}$$

This **cut-off property** is why the PACF helps select the AR order.

Population vs. Sample PACF, and Confidence Bands

ϕ_{nn} is a population quantity; in practice we estimate $\hat{\phi}_{11}, \hat{\phi}_{22}, \ldots$ from a finite sample – the **sample PACF**. Even for a true AR(p), sampling noise means $\hat{\phi}_{kk} \neq 0$ exactly for $k > p$; we instead look for coefficients that are **statistically significant**, using the approximate band

$$\boxed{\pm \frac{1.96}{\sqrt{T}}}$$

A coefficient outside the band is interpreted as significantly non-zero. For an AR(p): significant PACF at lags $1, \ldots, p$, then ≈ 0 afterward.

ACF versus PACF

	ACF	**PACF**
Measures	$\rho_k = \text{Corr}(X_t, X_{t-k})$	correlation controlling for lags $1, \ldots, k-1$
Contains	direct *and* indirect effects	incremental linear association only
AR(p) pattern	decays gradually	cuts off exactly after lag p

Order Selection via AIC and BIC

For each candidate order $p = 0, 1, \ldots, p_{\max}$, fit an AR($p$) and compute

$$\boxed{AIC = -2\ell(\hat{\theta}) + 2k} \qquad \boxed{BIC = -2\ell(\hat{\theta}) + k \log T}$$

where $\ell(\hat{\theta})$ is the maximized log-likelihood and k the number of parameters. Both trade off **fit vs. complexity**; select

$$\boxed{\hat{p}_{AIC} = \arg\min_p AIC(p)} \qquad \boxed{\hat{p}_{BIC} = \arg\min_p BIC(p)}$$

The BIC penalty $k \log T$ grows faster with T than AIC's $2k$, so BIC favors more parsimonious models in large samples.

Order Selection: the Full Logic

Observe $X_1, \ldots, X_T$

$\Downarrow$

True AR order p unknown

$\Downarrow$

PACF(k) = incremental contribution of lag k

$\Downarrow$

For AR(p) : PACF cuts off after p

$\Downarrow$

Use sample PACF and/or AIC/BIC to select a plausible p

A Worked AR(1) Fit (AutoReg, $T = 3938$)

Estimated model: $\boxed{X_t = 0.0646 - 0.0889\, X_{t-1} + \varepsilon_t}$, $\hat{\sigma}_\varepsilon = 1.738$.

$\hat{\phi}_1 = -0.0889$, $SE = 0.016$, $z = -0.0889/0.016 \approx -5.602$, $p < 0.001 \Rightarrow$ reject $H_0 : \phi_1 = 0$.

The 95% CI $[-0.120, -0.058]$ excludes zero.

$\boxed{\text{statistical significance} \neq \text{large economic magnitude}}$ $(|\hat{\phi}_1| = 0.0889$ is small)

The Constant Is Not the Mean

The constant $\hat{\phi}_0 = 0.0646$ is itself statistically significant ($z = 2.330$, $p = 0.020$), but in an AR(1) it is **not** the unconditional mean. Taking expectations of $X_t = \phi_0 + \phi_1 X_{t-1} + \varepsilon_t$: $\mu = \phi_0 + \phi_1 \mu \Rightarrow \mu(1 - \phi_1) = \phi_0$, so

$$\boxed{\mu = \frac{\phi_0}{1 - \phi_1}}$$

Numerically: $\hat{\mu} = \dfrac{0.0646}{1 - (-0.0889)} = \dfrac{0.0646}{1.0889} \approx \boxed{0.059}$.

Stationarity Check and Shock Decay

$|\hat{\phi}_1| = 0.0889 < 1$: the fitted dynamics are stationary. Effect of a shock after h periods $\propto \phi_1^h$:

$$\phi_1^1 = -0.0889, \qquad \phi_1^2 \approx 0.0079, \qquad \phi_1^3 \approx -0.0007$$

The sign alternates (since $\phi_1 < 0$) but decays almost immediately ($|\phi_1| \ll 1$):
shocks are highly transitory and disappear rapidly.

The AR Root

The AR polynomial $1 - \phi_1 z = 0$ has root $z = 1/\phi_1 = 1/(-0.0889) \approx -11.25$, matching the reported root $z = -11.2472$, with $|z| = 11.2472$. Stationarity requires roots **outside** the unit circle, $|z| > 1$ – satisfied here. The two stationarity checks are equivalent:

$$\boxed{|\phi_1| < 1 \iff \left|\frac{1}{\phi_1}\right| > 1}$$

$\hat{\sigma}_\varepsilon = 1.738$ measures the unpredictable component left after the AR(1) dynamics. Reported: $\ell(\hat{\theta}) = -7761.702$, $AIC = 15529.404$, $BIC = 15548.239$, $HQIC = 15536.086$.

Important

A single AIC value is **not** meaningful in isolation – it is useful only when **compared** across candidate models (e.g. AR(1), AR(2), AR(3), ...), always choosing $\arg\min_p AIC(p)$.

ARIMA(1,0,0) $\equiv$ AR(1)

With $d = 0$ (no differencing) and $q = 0$ (no MA part): $\boxed{\text{ARIMA}(1,0,0) \equiv \text{AR}(1)}$. Estimated model ($T = 3938$):

$$\boxed{X_t = 0.0597 - 0.0889\, X_{t-1} + \varepsilon_t}, \qquad \hat{\sigma}_\varepsilon^2 = 3.0196 \;\Rightarrow\; \hat{\sigma}_\varepsilon = \sqrt{3.0196} \approx 1.738$$

AR Coefficient and Conditional Expectation

$\hat{\phi}_1 = -0.0889$, $SE = 0.007$, $z = -11.859$, $p < 0.001$; 95% CI $[-0.104, -0.074]$. Assuming $\mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0$, the one-step conditional expectation is

$$\boxed{\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] = 0.0597 - 0.0889\, X_{t-1}}$$

Again, highly significant but economically small:
statistical significance $\neq$ large predictive magnitude.

The statsmodels ARIMA Constant Is the Mean

ARIMA parameterizes around the long-run mean μ:

$$\boxed{X_t - \mu = \phi_1(X_{t-1} - \mu) + \varepsilon_t}$$

so the reported const is $\hat{\mu} = 0.0597$ – do **not** divide by $1 - \phi_1$ again. Expanding: $X_t - \mu = \phi_1 X_{t-1} - \phi_1\mu + \varepsilon_t \Rightarrow X_t = \mu(1 - \phi_1) + \phi_1 X_{t-1} + \varepsilon_t$, giving the conventional-form intercept

$$\boxed{c = \mu(1 - \phi_1)}$$

Checking the Two Parameterizations Agree

Using $\hat{\mu} = 0.0597$, $\hat{\phi}_1 = -0.0889$:

$$\hat{c} = 0.0597\big(1 - (-0.0889)\big) \approx \boxed{0.0650}$$

The two equivalent fitted forms:

$$\boxed{X_t - 0.0597 = -0.0889(X_{t-1} - 0.0597) + \varepsilon_t}$$

$$\iff \qquad \boxed{X_t \approx 0.0650 - 0.0889\, X_{t-1} + \varepsilon_t}$$

So $\underbrace{0.0597}_{\texttt{const in ARIMA}} = \mu$, while $\underbrace{0.0650}_{c} = \mu(1 - \phi_1)$ – the same word "constant" means different things in different parameterizations.

$|\hat{\phi}_1| = 0.0889 < 1$: stationary. Shock decay: $\hat{\phi}_1^1 = -0.0889$, $\hat{\phi}_1^2 = 0.0079$, $\hat{\phi}_1^3 = -0.0007$ – alternating sign, rapid decay.
Innovation variance $\hat{\sigma}^2 = 3.0196$ (95% CI $[2.960, 3.080]$) $\Rightarrow \hat{\sigma} \approx 1.738$. Reported: $\ell(\hat{\theta}) = -7763.745$, $AIC = 15533.491$, $BIC = 15552.326$, $HQIC = 15540.173$ – again only meaningful **relative** to competing orders.

Ljung–Box Test for Residual Autocorrelation

A well-specified model should leave residuals that behave like unpredictable innovations: Residuals $\approx$ unpredictable shocks.

H_0 : no residual autocorrelation up to the tested lag $\qquad$ H_1 : autocorrelation remains

Example: $Q(1) = 0.12$, $p = 0.73 > 0.05 \Rightarrow$ fail to reject H_0 – no evidence of remaining lag-1 autocorrelation (the AR(1) term captured the linear dependence in the mean).

Jarque–Bera Test, Skewness and Kurtosis

H_0 : residuals are normal $\qquad$ H_1 : residuals are not normal

Example: $JB = 12660.48$, $p < 0.001 \Rightarrow$ strongly reject normality. Decomposing why: Skew $= -0.16$ (mild left asymmetry vs. 0 for Gaussian); Kurtosis $= 11.78 \gg 3$ (the Gaussian benchmark) $\Rightarrow$ heavy tails, far more extreme observations than Gaussian – the main driver of the large JB statistic, and typical of financial returns.

Heteroskedasticity: Mean vs. Variance

Example heteroskedasticity statistic $H = 0.85$, $p < 0.001 \Rightarrow$ reject constant residual variance.

$$\boxed{\text{modeling the conditional mean} \neq \text{modeling the conditional variance}}$$

The AR(1) specifies $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ but says nothing about $\text{Var}(X_t \mid \mathcal{F}_{t-1})$. This motivates ARCH/GARCH models of conditional volatility.

Are the Residuals White Noise?

Ljung–Box $\Rightarrow \text{Corr}(\hat{\varepsilon}_t, \hat{\varepsilon}_{t-1}) \approx 0$ (no linear autocorrelation), **but** Jarque–Bera and the heteroskedasticity test both reject at $p < 0.001$. So:

little remaining first-order linear autocorrelation but non-normality and non-constant variance

Residuals are therefore **not** i.i.d. Gaussian white noise – checking autocorrelation alone is not a sufficient diagnostic.

Same Dynamics, Different Numbers

`AutoReg` and `ARIMA(1,0,0)` both fit the same AR(1), giving $\hat{\phi}_1 \approx -0.0889$ in both cases – yet their log-likelihoods, AICs, and BICs differ. Why? The two implementations build the likelihood **differently**, mainly in how they treat the parameterization of the constant and the first observation.

Different Parameterizations of the Constant

`AutoReg`: $\boxed{X_t = c + \phi_1 X_{t-1} + \varepsilon_t}$, giving $\hat{c} = 0.0646$. `ARIMA`: $\boxed{X_t - \mu = \phi_1(X_{t-1} - \mu) + \varepsilon_t}$, giving $\hat{\mu} = 0.0597$. Expanding the ARIMA form: $X_t = \mu(1 - \phi_1) + \phi_1 X_{t-1} + \varepsilon_t$, so

$$\boxed{c = \mu(1 - \phi_1)}$$

Numerically $\hat{c} = 0.0597[1 - (-0.0889)] \approx 0.0650 \approx 0.0646$: the apparent gap is mostly just a reparameterization, not a different model.

AutoReg: Conditional Likelihood

Modeling X_1 via the AR equation would need an unavailable X_0. `AutoReg` instead **conditions** on the needed initial observation(s) and builds the likelihood from what remains:

$$L_{\text{conditional}} = \prod_{t=2}^{T} f(X_t \mid X_{t-1}) \qquad \ell_{\text{conditional}} = \sum_{t=2}^{T} \log f(X_t \mid X_{t-1})$$

This is why `AutoReg` reports "Method: Conditional MLE."

ARIMA: State-Space Likelihood

`statsmodels`' `ARIMA` uses state-space (SARIMAX) machinery, initializing from the **stationary distribution**: for a stationary Gaussian AR(1), $\mathrm{Var}(X_t) = \sigma_\varepsilon^2/(1-\phi_1^2)$, so conceptually $X_1 \sim \mathcal{N}(\mu,\, \sigma_\varepsilon^2/(1-\phi_1^2))$, giving

$$\boxed{L_{\text{ARIMA}} = f(X_1)\prod_{t=2}^{T} f(X_t \mid X_{t-1})}$$

rather than simply conditioning on X_1. The two likelihoods are therefore **not** numerically identical.

Consequences: Likelihood, and Why Coefficients Still Match

Reported: $\ell_{\text{AutoReg}} = -7761.702$ vs. $\ell_{\text{ARIMA}} = -7763.745$ – different, even though both estimate $\hat{\phi}_1 \approx -0.0889$. With $T = 3938$ observations, how one or two initial points are treated barely moves the coefficient estimate, but it does shift the likelihood value itself:

$$\boxed{\text{same underlying AR dynamics} \not\Rightarrow \text{same likelihood value}}$$

Consequences for AIC and BIC

With $k = 3$ parameters $(\mu, \phi_1, \sigma_\varepsilon^2)$ and $\ell(\hat{\theta}) = -7763.745$:
$AIC = -2(-7763.745) + 2(3) = 15527.490 + 6 = 15533.490$ – matching the reported ARIMA value 15533.491. Since $AIC_{\text{AutoReg}} = 15529.404 \neq AIC_{\text{ARIMA}} = 15533.491$ (and similarly for BIC: 15548.239 vs. 15552.326):

$$\boxed{\text{different likelihood} \Longrightarrow \text{different AIC/BIC}}$$

A Caution and Its Resolution

$$AIC_{\text{AutoReg}} < AIC_{\text{ARIMA}} \not\Rightarrow \texttt{AutoReg is a better AR(1) model}$$

AIC/BIC comparisons are meaningful only when models share the same data, the same likelihood definition, the same treatment of initial observations, and comparable distributional assumptions. **Rule:** to select an order p, compare $AIC(1), AIC(2), \ldots$ *within one consistent framework* (e.g. all via `ARIMA`, or all via `AutoReg`) – never across the two.

Summary: AutoReg vs. ARIMA(1,0,0)

	AutoReg	ARIMA(1,0,0)
Dynamics	AR(1)	AR(1)
AR coefficient	≈ -0.0889	≈ -0.0889
Reported constant	intercept c	long-run mean μ
Relation	$c = \mu(1-\phi_1)$	
Likelihood	Conditional	State-space
Log-likelihood	-7761.702	-7763.745
AIC / BIC	15529.404 / 15548.239	15533.491 / 15552.326

same AR(1) dynamics, different conventions $\Rightarrow$ similar $\hat{\phi}_1$, different ℓ, *AIC*, *BIC*

Normalized AIC and BIC

Dividing the standard criteria by sample size T:

$$\boxed{AIC_{\text{norm}} = -\frac{2}{T}\ln L(\hat{\theta}) + \frac{2k}{T} = \frac{AIC_{\text{standard}}}{T}}$$

$$\boxed{BIC_{\text{norm}} = -\frac{2}{T}\ln L(\hat{\theta}) + \frac{k\ln T}{T} = \frac{BIC_{\text{standard}}}{T}}$$

The two conventions differ only by the positive scale factor $1/T$ (same T for all candidates).

Why Normalization Doesn't Change Model Selection

Since $T > 0$, dividing every candidate's criterion by the same constant preserves their ordering: $AIC(p_1) < AIC(p_2) \iff AIC(p_1)/T < AIC(p_2)/T$. Hence

$$\arg\min_p AIC(p) = \arg\min_p \frac{AIC(p)}{T} \qquad \arg\min_p BIC(p) = \arg\min_p \frac{BIC(p)}{T}$$

Normalizing by T never changes which model is selected.

Step 1: Density of One Innovation

Assume $\boxed{\varepsilon_t \overset{iid}{\sim} \mathcal{N}(0, \sigma^2)}$. Since $Z \sim \mathcal{N}(\mu, \sigma^2)$ has density $f_Z(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\big(-\frac{(z-\mu)^2}{2\sigma^2}\big)$, with $\mu = 0$:

$$\boxed{f_\varepsilon(\varepsilon_t) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-1/2} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2}\right)}$$

Step 2: Innovations Are Prediction Errors

From $X_t = \phi_0 + \sum_{j=1}^{p} \phi_j X_{t-j} + \varepsilon_t$, isolate

$$\boxed{\varepsilon_t = X_t - \phi_0 - \sum_{j=1}^{p} \phi_j X_{t-j} = X_t - \mathbb{E}[X_t \mid X_{t-1}, \ldots, X_{t-p}]}$$

Conditioning on the past, define $a_t = \phi_0 + \sum_j \phi_j X_{t-j}$, a known constant given the past, so $X_t = a_t + \varepsilon_t$.

Step 2 (cont'd): Shifting a Gaussian

If $Z \sim \mathcal{N}(0, \sigma^2)$, then for any constant a: $\boxed{a + Z \sim \mathcal{N}(a, \sigma^2)}$ – adding a constant shifts the mean but leaves the variance unchanged. Applying this to $X_t = a_t + \varepsilon_t$:

$$\boxed{X_t \mid X_{t-1}, \ldots, X_{t-p} \;\sim\; \mathcal{N}\left(\phi_0 + \sum_{j=1}^{p} \phi_j X_{t-j},\; \sigma^2\right)}$$

Step 3: Conditional Density of X_t

$$f(X_t \mid X_{t-1}, \ldots, X_{t-p}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{\left(X_t - \phi_0 - \sum_{j=1}^{p} \phi_j X_{t-j}\right)^2}{2\sigma^2} \right]$$

Since the bracket is exactly ε_t:

$$\boxed{f(X_t \mid X_{t-1}, \ldots, X_{t-p}; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{\varepsilon_t^2}{2\sigma^2} \right)}$$

Step 4: Build the Joint Likelihood

Because the ε_t are i.i.d., the joint likelihood multiplies the individual densities: $L(\theta) = \prod_{t=1}^{T} f_\varepsilon(\varepsilon_t)$. Separating the two factors in the product and using $\prod_t e^{a_t} = e^{\sum_t a_t}$:

$$\boxed{L(\theta) = (2\pi\sigma^2)^{-T/2} \exp\left(-\frac{1}{2\sigma^2} \sum_{t=1}^{T} \varepsilon_t^2 \right)}$$

Step 5: Take the Logarithm

$\ell(\theta) = \ln L(\theta)$. Using $\ln(ab) = \ln a + \ln b$, $\ln(a^b) = b \ln a$, $\ln(e^x) = x$:

$$\ell(\theta) = \ln\left[(2\pi\sigma^2)^{-T/2}\right] + \ln\left[\exp\left(-\tfrac{1}{2\sigma^2}\textstyle\sum_t \varepsilon_t^2\right)\right] = -\frac{T}{2}\ln(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}\varepsilon_t^2$$

Step 6: Expand and Plug in the MLE of σ^2

Since $\ln(2\pi\sigma^2) = \ln(2\pi) + \ln(\sigma^2)$:

$$\ell(\theta) = -\frac{T}{2}\ln(2\pi) - \frac{T}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{t=1}^{T}\varepsilon_t^2$$

At the MLE, $\boxed{\hat{\sigma}^2 = \frac{1}{T}\sum_{t=1}^{T}\hat{\varepsilon}_t^2}$, so $\sum_t \hat{\varepsilon}_t^2 = T\hat{\sigma}^2$. Substituting:

$$\ell(\hat{\theta}) = -\frac{T}{2}\big[\ln(2\pi) + \ln(\hat{\sigma}^2) + 1\big] \quad \Longrightarrow \quad \boxed{-\frac{2}{T}\ell(\hat{\theta}) = \ln(\hat{\sigma}^2) + \underbrace{\ln(2\pi) + 1}_{\text{constant}}}$$

The Simplified AIC and BIC Formulas

Substituting into the normalized $AIC = -\frac{2}{T}\ell(\hat{\theta}) + \frac{2k}{T}$:

$$AIC = \ln(\hat{\sigma}^2) + \frac{2k}{T} + \underbrace{\ln(2\pi) + 1}_{\text{same for every candidate model}}$$

Dropping the model-independent constant (it never affects $\arg\min$):

$$\boxed{AIC(k) = \ln \hat{\sigma}_k^2 + \frac{2k}{T}} \qquad \boxed{BIC(k) = \ln \hat{\sigma}_k^2 + \frac{k \ln T}{T}}$$

Interpretation: Fit versus Complexity

$$AIC(k) = \underbrace{\ln \hat{\sigma}_k^2}_{\text{goodness of fit}} + \underbrace{\frac{2k}{T}}_{\text{complexity penalty}}$$

$$BIC(k) = \underbrace{\ln \hat{\sigma}_k^2}_{\text{goodness of fit}} + \underbrace{\frac{k \ln T}{T}}_{\text{complexity penalty}}$$

More lags $\Rightarrow$ lower $\hat{\sigma}_k^2$ (better fit) but more parameters (bigger penalty). Since the AIC penalty per parameter is $2/T$ and BIC's is $(\ln T)/T$: whenever $\ln T > 2$ (i.e. $T > e^2 \approx 7.39$ – true for virtually all real samples), $\boxed{\frac{\ln T}{T} > \frac{2}{T}}$, so BIC penalizes complexity more and favors sparser models.

A Note on Counting Parameters, and the `statsmodels` Convention

Counting only the mean parameters $\phi_0, \ldots, \phi_p$ gives $k = p + 1$; including σ^2 gives $k = p + 2$. Different sources use different conventions – just stay consistent across candidate models. The `statsmodels` package reports the **unnormalized** forms $\boxed{AIC = -2\ell(\hat{\theta}) + 2k}$, $\boxed{BIC = -2\ell(\hat{\theta}) + k \ln T}$ – numerically different from $\ln \hat{\sigma}^2 + 2k/T$, but encoding the same fit-vs-complexity trade-off.

The Full Derivation, at a Glance

$$AIC = -2\ell(\hat{\theta}) + 2k$$

$$\Downarrow \text{ divide by } T$$

$$-\tfrac{2}{T}\ell(\hat{\theta}) + \tfrac{2k}{T}$$

$$\Downarrow \text{ Gaussian AR likelihood}$$

$$\ln(\hat{\sigma}^2) + \ln(2\pi) + 1 + \tfrac{2k}{T}$$

$$\Downarrow \text{ drop constants common to all models}$$

$$\boxed{AIC(k) = \ln \hat{\sigma}_k^2 + \tfrac{2k}{T}}$$

(and identically for $BIC(k) = \ln \hat{\sigma}_k^2 + \frac{k \ln T}{T}$, replacing $2k/T$ with $k \ln T/T$ throughout).

Conditional Heteroskedasticity: Constant vs. Clustered Variance

After estimating an ARMA model, we must check whether the residuals $\widehat{\varepsilon}_t$ exhibit **conditional heteroskedasticity**.

Homoskedastic benchmark

$$\boxed{\mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma^2}$$

where $\mathcal{F}_{t-1}$ is all information available up to $t-1$: knowing the past does not help predict whether volatility will be high or low today.

Financial returns routinely violate this: large moves tend to be followed by large moves, calm periods by calm periods – **volatility clustering**.

Volatility Clustering: A Numerical Illustration

Consider the residual sequence

$$0.1,\ -0.2,\ 0.1,\ 2.5,\ -3.0,\ 2.1,\ -0.1,\ 0.2.$$

Signs alternate, so there may be little visible autocorrelation in the residuals themselves. Their squares,

$$0.01,\ 0.04,\ 0.01,\ 6.25,\ 9.00,\ 4.41,\ 0.01,\ 0.04,$$

show a clear cluster of large values. Hence it is entirely possible to have

$$\boxed{\mathrm{Corr}(\varepsilon_t, \varepsilon_{t-1}) \approx 0} \qquad \text{while} \qquad \boxed{\mathrm{Corr}(\varepsilon_t^2, \varepsilon_{t-1}^2) > 0.}$$

Why Study Squared Residuals?

Volatility measures the **magnitude** of fluctuations, not their direction: $\varepsilon_t = 3$ and $\varepsilon_t = -3$ are opposite in sign but identical in magnitude, and

$$3^2 = (-3)^2 = 9.$$

$$\boxed{\varepsilon_t^2}$$

is therefore a simple proxy for the magnitude of the shock – and hence for volatility. Engle's ARCH test asks: *do past squared residuals help predict the current squared residual?*

The ARCH Auxiliary Regression

$$\widehat{\varepsilon}_t^2 = \alpha_0 + \alpha_1 \widehat{\varepsilon}_{t-1}^2 + \alpha_2 \widehat{\varepsilon}_{t-2}^2 + \cdots + \alpha_q \widehat{\varepsilon}_{t-q}^2 + u_t$$

- Dependent variable: the current squared residual $\widehat{\varepsilon}_t^2$.
- Regressors: q lagged squared residuals $\widehat{\varepsilon}_{t-1}^2, \ldots, \widehat{\varepsilon}_{t-q}^2$.

Hypotheses

$H_0 : \alpha_1 = \cdots = \alpha_q = 0$ (no ARCH effects)

$H_1 : \exists j,\ \alpha_j \neq 0$ (ARCH effects present)

Engle's Lagrange Multiplier Statistic

Estimate the auxiliary regression $\widehat{\varepsilon}_t^2 = \alpha_0 + \sum_{j=1}^{q} \alpha_j \widehat{\varepsilon}_{t-j}^2 + u_t$ and record its R^2. Engle's ARCH LM statistic is

$$\boxed{LM = TR^2}$$

where T is the number of observations in the auxiliary regression. Under H_0,

$$\boxed{LM \overset{a}{\sim} \chi_q^2.}$$

Decision rule (level 5%)

p-value $< 0.05 \Rightarrow$ evidence of ARCH effects. p-value $\geq 0.05 \Rightarrow$ insufficient evidence – this does *not* prove the variance is exactly constant, only that no ARCH dependence was detected at the lags tested.

Restricted vs. Unrestricted Auxiliary Regressions

The F-statistic tests the same H_0 as the LM statistic, by comparing two nested models.

Restricted model (H_0 imposed)

$$\boxed{\widehat{\varepsilon}_t^2 = \alpha_0 + u_t}$$

Unrestricted model

$$\boxed{\widehat{\varepsilon}_t^2 = \alpha_0 + \sum_{j=1}^{q} \alpha_j \widehat{\varepsilon}_{t-j}^2 + u_t}$$

Since the unrestricted model nests the restricted one, $\boxed{SSR_U \leq SSR_R}$: the question is whether the improvement in fit $SSR_R - SSR_U$ is *statistically* significant, not merely non-negative.

Constructing the F-Statistic

With n observations and q lagged squared residuals:

$$\boxed{F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n-q-1)}}$$

- Numerator $\dfrac{SSR_R - SSR_U}{q}$: average improvement in fit *per relaxed restriction* (there are q restrictions $\alpha_1 = 0, \ldots, \alpha_q = 0$).
- Denominator $\dfrac{SSR_U}{n-q-1}$: residual variance of the unrestricted regression, which estimates $q+1$ parameters, leaving $n-q-1$ residual degrees of freedom.

$$\boxed{F = \frac{\text{improvement in fit per additional ARCH parameter}}{\text{remaining unexplained variation per residual d.o.f.}}}$$

Intuition and the Reference Distribution

- If H_0 holds, adding the lagged squared residuals barely helps: $SSR_R \approx SSR_U \Rightarrow F$ small.
- If past squared residuals strongly predict the current one: $SSR_U \ll SSR_R \Rightarrow F$ large.

$$\boxed{\text{large } F \implies \text{lagged squared residuals have substantial joint explanatory power}}$$

Under the usual assumptions and H_0,

$$\boxed{F \sim F_{q,\, n-q-1}}$$

with q = number of restrictions tested, and $n - q - 1$ = residual d.o.f. of the unrestricted regression. At the 5% level: `f_pvalue` $< 0.05 \Rightarrow$ reject H_0 (ARCH effects present).

LM Test vs. F-Test: Same Question, Different Statistics

Both tests target $H_0 : \alpha_1 = \cdots = \alpha_q = 0$, but build their statistic differently.

Test	Statistic	Reference distribution
Engle LM	nR^2	χ^2_q
F-test	$\dfrac{(SSR_R - SSR_U)/q}{SSR_U/(n-q-1)}$	$F_{q,\,n-q-1}$

Both ask: do past squared shocks contain information about current volatility?

In large samples the two tests generally agree, although their exact statistics and p-values differ numerically.

Decomposing a Return Series: Mean and Variance

A return series decomposes as

$$\boxed{r_t = \mu_t + \varepsilon_t}, \qquad \boxed{\mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}]} \text{ (conditional mean)}$$

The innovation itself is $\boxed{\varepsilon_t = \sigma_t z_t}$ with $\mathbb{E}[z_t] = 0$, $\mathrm{Var}(z_t) = 1$, hence $\boxed{\mathrm{Var}(\varepsilon_t \mid \mathcal{F}_{t-1}) = \sigma_t^2}$.

$$\boxed{\begin{array}{rc} \text{Conditional mean:} & \mu_t = \mathbb{E}[r_t \mid \mathcal{F}_{t-1}] \\ \text{Conditional variance:} & \sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) \end{array}}$$

These are **two separate modelling choices**: the mean specification answers the first equation, the GARCH order answers the second.

Zero, Constant, and AR Conditional Means

Zero mean

$\boxed{\mu_t = 0} \Rightarrow r_t = \varepsilon_t = \sigma_t z_t$. No predictable component: $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$. All GARCH dynamics concern the variance only.

Constant mean

$\boxed{\mu_t = \mu} \Rightarrow r_t = \mu + \varepsilon_t$. Now μ is an additional estimated parameter, with $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mu$.

AR(1) mean

$\boxed{\mu_t = \mu + \phi_1 r_{t-1}} \Rightarrow r_t = \mu + \phi_1 r_{t-1} + \varepsilon_t$, i.e. $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mu + \phi_1 r_{t-1}$ – an AR(1)–GARCH(1,1) model combined with $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$.

Don't Confuse Mean Lags with GARCH Orders

In an AR-mean GARCH specification, two *separate* order parameters appear:

`lags`	$\longrightarrow$	autoregressive order of the *conditional mean*
p	$\longrightarrow$	lags of squared innovations in the variance eq.
q	$\longrightarrow$	lags of conditional variance in the variance eq.

`lags=1` gives $r_t = \mu + \phi_1 r_{t-1} + \varepsilon_t$; p=1, q=1 instead govern $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2$ – two independent modelling decisions.

Summary of Mean Specifications

$\mu_t = 0$	$\longrightarrow$	Zero mean
$\mu_t = \mu$	$\longrightarrow$	Constant mean
$\mu_t = \mu + \sum_{i=1}^{P} \phi_i r_{t-i}$	$\longrightarrow$	AR(P) mean

"mean" answers: how should $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]$ be modeled?

"vol" answers: how should $\text{Var}(r_t \mid \mathcal{F}_{t-1})$ be modeled?

Aside: Whitespace Delimiters When Reading Data

A technical parsing note: specifying a column delimiter as the regular expression \s+ means **one or more consecutive whitespace characters**.

\s = any whitespace character (space, tab, newline, ...)

\+ = one or more occurrences of the preceding expression

Hence \s+ matches " ", " ", or a tab: the number of spaces between two values need not be constant.

Testing Joint Explanatory Power

For $\boxed{Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i}$, the regression's F-statistic tests whether the explanatory variables have **joint** explanatory power.

$$\boxed{H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0} \qquad \boxed{H_1 : \text{at least one } \beta_j \neq 0}$$

Note β_0 is *excluded* from the hypothesis. Under H_0: $\boxed{Y_i = \beta_0 + \varepsilon_i}$, so the fitted value is simply $\widehat{Y}_i = \overline{Y}$: do the regressors jointly improve on predicting Y by its mean alone?

Restricted/Unrestricted Models and the F-Statistic

Restricted: $\boxed{Y_i = \beta_0 + \varepsilon_i}$. **Unrestricted:** $\boxed{Y_i = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik} + \varepsilon_i}$.
Since the unrestricted model nests the restricted one, $SSR_U \leq SSR_R$ always; the real question is whether the improvement is significant.

$$\boxed{F = \frac{(SSR_R - SSR_U)/k}{SSR_U/(n-k-1)}}$$

with k = number of slopes, n = sample size. Numerator: average fit improvement per added regressor. Denominator: residual variance per d.o.f.

Intuition and the R^2-Based Formula

If H_0 is true, adding regressors barely helps: $SSR_R \approx SSR_U \Rightarrow F$ small. If the regressors carry real information, $SSR_U \ll SSR_R \Rightarrow F$ large:

$$\boxed{\text{large } F \implies \text{strong evidence against } H_0}$$

With an intercept, F has an equivalent R^2-based form:

$$\boxed{F = \frac{R^2/k}{(1-R^2)/(n-k-1)}}$$

numerator = explained variation per regressor; denominator = unexplained variation per residual d.o.f.

Worked Example: The Housing Regression

With $n = 401$, $k = 13$, $R^2 = 0.735$:

$$F = \frac{0.735/13}{(1 - 0.735)/(401 - 13 - 1)}, \qquad \text{residual d.o.f.} = 401 - 13 - 1 = 387 = Df_{\text{Residuals}}$$

$$F = \frac{0.735/13}{0.265/387} \approx 82.6 \approx \boxed{F = 82.66}$$

Under H_0: $\boxed{F \sim F_{13,387}}$, and $\boxed{\text{Prob}(F\text{-stat}) = P(F_{13,387} \geq 82.66 \mid H_0) = 5.72 \times 10^{-103}}$. Since this is $\ll 0.05$, we strongly reject $H_0 : \beta_1 = \cdots = \beta_{13} = 0$: the regressors are jointly significant.

Global F-Test vs. Individual t-Tests

t-test	$\longrightarrow$	is one particular β_j significant?
F-test	$\longrightarrow$	are the regressors *jointly* significant?

A t-test is $H_0 : \beta_j = 0$ vs. $H_1 : \beta_j \neq 0$ (conditional on the other regressors); the F-test is $H_0 : \beta_1 = \cdots = \beta_k = 0$ (all at once).

Both can disagree

A highly significant global F-test can coexist with some individual $p_j > 0.05$: the F-test only says the coefficients are not *all simultaneously* zero, not that each one individually matters.

Testing Normality via Skewness and Kurtosis

Both the Omnibus and Jarque–Bera tests target

H_0 : residuals are normally distributed $\qquad$ H_1 : residuals are not normal

For a normal distribution, skewness $S = 0$ and kurtosis $K = 3$. The tests differ in **how** S and K are combined into a statistic.

The Jarque–Bera Statistic

$$\boxed{JB = \frac{n}{6}\left[S^2 + \frac{(K-3)^2}{4}\right]}$$

$$\boxed{JB = \frac{n}{6}\Big[\underbrace{S^2}_{\text{departure from symmetry}} + \underbrace{\frac{(K-3)^2}{4}}_{\text{departure from normal kurtosis}}\Big]}$$

$K-3$ is the **excess kurtosis**. For a Gaussian, $S=0$ and $K-3=0$, so JB is small; large skewness/kurtosis deviations inflate it. Under H_0: $\boxed{JB \overset{a}{\sim} \chi^2_2}$.

The Omnibus (D'Agostino) Test

`statsmodels`' Omnibus statistic is D'Agostino's K^2 test. It first transforms sample skewness/kurtosis into statistics Z_S, Z_K that are approximately $\mathcal{N}(0,1)$ under normality, then combines them:

$$\boxed{K^2 = Z_S^2 + Z_K^2} \qquad \boxed{K^2 \overset{a}{\sim} \chi^2_2 \text{ under } H_0}$$

These transformations better account for the finite-sample behavior of skewness and kurtosis estimators.

Jarque–Bera	$\rightarrow$	uses S, $K-3$ directly
Omnibus	$\rightarrow$	transforms S, K first, then combines

Applying Both Tests to a Regression Output

Observed: $\boxed{S = 1.189}$, $\boxed{K = 8.128}$ (vs. 0 and 3 under normality) – substantial positive skew and much heavier tails than Gaussian.

$$\boxed{\text{Omnibus} = 117.115,\ \text{Prob(Omnibus)} \approx 0} \qquad \boxed{JB = 533.859,\ \text{Prob}(JB) = 1.19 \times 10^{-116}}$$

Both p-values are far below 5%: strong evidence that the residuals are **not** normal.

Definition

The DW statistic detects **first-order serial correlation** in residuals $\widehat{\varepsilon}_1, \ldots, \widehat{\varepsilon}_T$:

$$DW = \frac{\sum_{t=2}^{T} (\widehat{\varepsilon}_t - \widehat{\varepsilon}_{t-1})^2}{\sum_{t=1}^{T} \widehat{\varepsilon}_t^2}$$

Numerator: squared differences between consecutive residuals. Denominator: overall residual magnitude. DW compares consecutive-residual variation to total residual variation.

Range and Interpretation

$$\boxed{0 \leq DW \leq 4},$$

$DW \approx 2$	$\Rightarrow$	little/no first-order autocorrelation
$DW < 2$	$\Rightarrow$	positive first-order autocorrelation
$DW > 2$	$\Rightarrow$	negative first-order autocorrelation

Why $DW < 2$ Means Positive Autocorrelation

If consecutive residuals are positively correlated, a positive residual tends to be followed by another positive one: $\widehat{\varepsilon}_t \approx \widehat{\varepsilon}_{t-1}$, so

$$\widehat{\varepsilon}_t - \widehat{\varepsilon}_{t-1} \approx 0 \quad \Rightarrow \quad (\widehat{\varepsilon}_t - \widehat{\varepsilon}_{t-1})^2 \text{ small}$$

The DW numerator shrinks, giving $\boxed{DW < 2}$: $\boxed{\text{positive autocorrelation} \Rightarrow DW < 2}$.

Why $DW > 2$ Means Negative Autocorrelation

If consecutive residuals are negatively correlated, $\widehat{\varepsilon}_t \approx -\widehat{\varepsilon}_{t-1}$. E.g. $\widehat{\varepsilon}_{t-1} = 2$, $\widehat{\varepsilon}_t = -2$:

$$\widehat{\varepsilon}_t - \widehat{\varepsilon}_{t-1} = -2 - 2 = -4 \quad \Rightarrow \quad (\widehat{\varepsilon}_t - \widehat{\varepsilon}_{t-1})^2 = 16$$

The numerator grows, giving $\boxed{DW > 2}$: $\boxed{\text{negative autocorrelation} \Rightarrow DW > 2}$.

Link to the Lag-1 Autocorrelation

Let $\widehat{\rho}_1 = \text{Corr}(\widehat{\varepsilon}_t, \widehat{\varepsilon}_{t-1})$. Approximately,

$$\boxed{DW \approx 2(1 - \widehat{\rho}_1)} \quad \Longleftrightarrow \quad \boxed{\widehat{\rho}_1 \approx 1 - \frac{DW}{2}}$$

$$\begin{aligned}
\widehat{\rho}_1 \approx 0 &\quad \Leftrightarrow \quad DW \approx 2 \\
\widehat{\rho}_1 \approx 1 &\quad \Leftrightarrow \quad DW \approx 0 \text{ (very strong positive AC)} \\
\widehat{\rho}_1 \approx -1 &\quad \Leftrightarrow \quad DW \approx 4 \text{ (very strong negative AC)}
\end{aligned}$$

Worked Example & Important Caveats

Observed $\boxed{DW = 1.124 < 2}$: suggests positive first-order serial correlation. Using $\widehat{\rho}_1 \approx 1 - DW/2$:

$$\widehat{\rho}_1 \approx 1 - \frac{1.124}{2} = 1 - 0.562 = \boxed{0.438}$$

Two caveats

(1) This $\widehat{\rho}_1$ is only an approximation, not a formal test. (2) The reported DW is a **test statistic**, not a p-value: a formal DW test needs critical bounds d_L, d_U depending on T, k, and the significance level.

Durbin–Watson vs. Other Residual Diagnostics

Omnibus / Jarque–Bera	$\rightarrow$	normality of residuals
Durbin–Watson	$\rightarrow$	first-order serial correlation
Ljung–Box	$\rightarrow$	serial correlation across several lags
ARCH test	$\rightarrow$	serial dependence in squared residuals

In short: $\boxed{\text{Corr}(\varepsilon_t, \varepsilon_{t-1})}$ concerns the residuals themselves, while $\boxed{\text{Corr}(\varepsilon_t^2, \varepsilon_{t-1}^2)}$ concerns their magnitude (ARCH effects) – these are genuinely different diagnostics.

Why the Design Matrix's Conditioning Matters

For $\boxed{\mathrm{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}}$ with $X \in \mathbb{R}^{n \times k}$, OLS gives

$$\boxed{\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathrm{y}}$$

provided X has full column rank. If some columns of X are exactly or nearly linear combinations of others, $X^\top X$ becomes singular or near-singular, creating numerical instability. The **condition number** quantifies this.

The Singular Value Decomposition

Any $X \in \mathbb{R}^{n \times k}$ $(n \geq k)$ decomposes as

$$\boxed{X = U \Sigma V^\top}, \qquad U \in \mathbb{R}^{n \times k},\ \Sigma \in \mathbb{R}^{k \times k},\ V \in \mathbb{R}^{k \times k}$$

with $\boxed{U^\top U = I_k,\ V^\top V = I_k}$ and Σ diagonal:

$$\boxed{\Sigma = \mathrm{diag}(\sigma_1, \ldots, \sigma_k)}, \qquad \boxed{\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0}$$

The σ_j are the **singular values** of X.

What Do Singular Values Measure?

For a vector $\mathrm{v} \in \mathbb{R}^k$, X maps it to $X\mathrm{v} \in \mathbb{R}^n$. The right singular vectors $\mathrm{v}_1, \ldots, \mathrm{v}_k$ are special orthogonal directions such that

$$\boxed{X\mathrm{v}_j = \sigma_j \mathrm{u}_j}$$

Since $\|\mathrm{u}_j\|_2 = \|\mathrm{v}_j\|_2 = 1$, we get $\boxed{\|X\mathrm{v}_j\|_2 = \sigma_j}$: σ_j is exactly how much X stretches direction v_j. A very small σ_j means some direction in regressor-space is almost collapsed to zero by X.

Geometric Picture of the SVD

$X = U\Sigma V^\top$ decomposes into three successive maps:

$$x \xrightarrow{V^\top} V^\top x \xrightarrow{\Sigma} \Sigma V^\top x \xrightarrow{U} U\Sigma V^\top x$$

$$X = \underbrace{U}_{\text{rotation}} \quad \underbrace{\Sigma}_{\text{stretching}} \quad \underbrace{V^\top}_{\text{rotation}}$$

$V^\top$ rotates/reflects the input coordinates; Σ stretches/compresses each direction by $\sigma_1, \ldots, \sigma_k$; U rotates the result into the output space.

Singular Values Are Square Roots of Eigenvalues of $X^\top X$

From $X = U\Sigma V^\top$: $X^\top = V\Sigma^\top U^\top$, so

$$X^\top X = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top\Sigma V^\top$$

using $U^\top U = I$. For the reduced SVD, $\Sigma^\top\Sigma = \Sigma^2$, so

$$\boxed{X^\top X = V\Sigma^2 V^\top}$$

– an eigendecomposition of $X^\top X$. Hence

$$\boxed{\lambda_j(X^\top X) = \sigma_j^2(X)} \qquad \Longleftrightarrow \qquad \boxed{\sigma_j(X) = \sqrt{\lambda_j(X^\top X)}}$$

Singular Values and (Near) Linear Dependence

Exact dependence: $\exists \mathrm{a} \neq 0$ with $\boxed{X\mathrm{a} = 0} \Rightarrow \boxed{\sigma_{\min}(X) = 0}$: X loses full column rank, $X^\top X$ is singular, $(X^\top X)^{-1}$ does not exist – **perfect multicollinearity**.
Near dependence: e.g. $X_3 \approx 2X_1 + 3X_2$, i.e. $2X_1 + 3X_2 - X_3 \approx 0$. For $\mathrm{a} = (2, 3, -1)^\top / \|\cdot\|$, $X\mathrm{a} \approx 0$, so

$$\boxed{\sigma_{\min}(X) \approx 0}$$

– exactly the poorly-conditioned case.

The Condition Number

Using the Euclidean matrix norm,

$$\kappa_2(X) = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)} = \frac{\sigma_1}{\sigma_k}$$

It compares the most strongly represented direction with the most weakly represented one.

$\sigma_{\min}$ comparable to $\sigma_{\max}$	$\Rightarrow$	$\kappa(X) \approx 1$ (well-conditioned)
$\sigma_{\min} \ll \sigma_{\max}$	$\Rightarrow$	$\kappa(X) \gg 1$ (ill-conditioned)
$\sigma_{\min} = 0$	$\Rightarrow$	$\kappa(X) = \infty$ (singular)

Why Conditioning Matters for OLS

Using the SVD, when X has full column rank,

$$\boxed{\widehat{\boldsymbol{\beta}} = V\Sigma^{-1}U^{\top}\mathrm{y}}, \qquad \Sigma^{-1} = \mathrm{diag}(1/\sigma_1, \ldots, 1/\sigma_k)$$

A tiny $\sigma_k \approx 0$ produces a huge $\boxed{1/\sigma_k \gg 1}$: small perturbations in the data get strongly **amplified** in $\widehat{\boldsymbol{\beta}}$. This is exactly why an ill-conditioned X yields unstable coefficient estimates.

Condition Number and the Variance of $\widehat{\boldsymbol{\beta}}$ (1/2)

Under homoskedasticity, $\text{Var}(\widehat{\boldsymbol{\beta}} \mid X) = \sigma_\varepsilon^2 (X^\top X)^{-1}$, and $X^\top X = V \Sigma^2 V^\top$. Using $(ABC)^{-1} = C^{-1} B^{-1} A^{-1}$:

$$(X^\top X)^{-1} = (V \Sigma^2 V^\top)^{-1} = (V^\top)^{-1} (\Sigma^2)^{-1} V^{-1}$$

Since V is orthogonal, $\boxed{V^{-1} = V^\top}$ and $\boxed{(V^\top)^{-1} = V}$. Since Σ is diagonal with $\sigma_j > 0$ (full column rank),

$$(\Sigma^2)^{-1} = \Sigma^{-2} = \text{diag}(1/\sigma_1^2, \ldots, 1/\sigma_k^2)$$

Condition Number and the Variance of $\widehat{\boldsymbol{\beta}}$ (2/2)

Putting the pieces together:

$$\boxed{(X^\top X)^{-1} = V\Sigma^{-2}V^\top}$$

so

$$\boxed{\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid X) = \sigma_\varepsilon^2\, V\Sigma^{-2}V^\top}$$

A small $\sigma_j \approx 0$ gives $\boxed{1/\sigma_j^2 \gg 1}$: near-linear dependence between regressors can produce very large variances (and standard errors) for some linear combinations of $\widehat{\boldsymbol{\beta}}$.

Connection with Multicollinearity

If $X_3 \approx 2X_1 + 3X_2$, the three regressors carry almost redundant information. Typical symptoms:

unstable coefficient estimates
large standard errors
small individual t-statistics
coefficients sensitive to small data changes

Yet the regressors may still be *jointly* useful: a significant global F-test can coexist with insignificant individual t-tests.

A Large Condition Number Need Not Mean Multicollinearity

The condition number is also sensitive to **scale**: if $X_1 \sim 10^{-2}$ while $X_2 \sim 10^{4}$, even weakly correlated regressors can produce very different singular values and hence a huge $\sigma_{\max}/\sigma_{\min}$.

large condition number $\not\Longrightarrow$ multicollinearity necessarily

it only means: the design matrix is ill-conditioned – the source must be investigated

Diagnosing the Source: Scaling, VIF

Standardize each regressor, $\boxed{Z_{ij} = (X_{ij} - \overline{X}_j)/s_j}$, so $\mathbb{E}[Z_j] \approx 0,\ \mathrm{Var}(Z_j) \approx 1$. If κ drops sharply after standardizing, scale differences were the main cause; if it stays large, this points to genuine near-linear dependence.

Variance Inflation Factor: regress X_j on the other regressors to get R_j^2, then

$$\boxed{VIF_j = \frac{1}{1 - R_j^2}}$$

$R_j^2 \approx 1 \Rightarrow VIF_j \gg 1$: X_j is almost fully explained by the others.

Condition number	$\rightarrow$	overall conditioning of X
VIF_j	$\rightarrow$	multicollinearity specific to X_j

Worked Example: Reading the "Cond. No." Warning

Output reports $\boxed{\text{Cond. No.} = 1.53 \times 10^4}$, i.e. $\boxed{\kappa(X) \approx 15{,}300}$ – large, triggering `statsmodels`' warning about possible multicollinearity or numerical problems. The word **might** matters: a large κ can stem from

$$\boxed{\text{strong multicollinearity} \quad \text{or} \quad \text{very different regressor scales} \quad \text{or both}}$$

Useful next steps: inspect variable scales, standardize and recompute κ, inspect pairwise correlations, and compute VIFs – before concluding multicollinearity is the culprit.

Why the Error Covariance Matters

The note *"Standard Errors assume the covariance matrix of the errors is correctly specified"* is about the validity of *SE*'s, *t*-stats, *p*-values, and confidence intervals – **not** necessarily a detected problem. All downstream inference rests on the assumed error covariance structure.

The Classical OLS Covariance Assumption

For $\boxed{y = X\beta + \varepsilon}$, the default (`Covariance Type: nonrobust`) assumption is

$$\boxed{\text{Var}(\varepsilon \mid X) = \sigma^2 I_n}$$

This bundles two separate assumptions:

1. Homoskedasticity

$\boxed{\text{Var}(\varepsilon_i \mid X) = \sigma^2 \ \forall i}$ – constant error variance across observations.

2. No cross-error correlation

$\boxed{\text{Cov}(\varepsilon_i, \varepsilon_j \mid X) = 0 \ (i \neq j)}$ – in particular, no serial correlation $\text{Cov}(\varepsilon_t, \varepsilon_{t-1} \mid X) = 0$ for time series.

Deriving the OLS Estimation Error

From $\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathrm{y}$ and $\mathrm{y} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\widehat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top (X\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (X^\top X)^{-1} X^\top X\boldsymbol{\beta} + (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$$

$$= \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}$$

Hence

$$\boxed{\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (X^\top X)^{-1} X^\top \boldsymbol{\varepsilon}}$$

– the estimation error is a *linear* function of the error vector $\boldsymbol{\varepsilon}$.

The General Covariance Formula

Conditional on X: $\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid X) = \mathrm{Var}[(X^\top X)^{-1} X^\top \boldsymbol{\varepsilon} \mid X]$. Using $\mathrm{Var}(A\mathrm{x}) = A\,\mathrm{Var}(\mathrm{x})A^\top$:

$$\boxed{\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \,\mathrm{Var}(\boldsymbol{\varepsilon} \mid X)\, X (X^\top X)^{-1}}$$

This is the **general** formula – valid *whatever* the true error covariance $\mathrm{Var}(\boldsymbol{\varepsilon} \mid X)$ turns out to be.

The Classical Homoskedastic Case

Substituting $\text{Var}(\varepsilon \mid X) = \sigma^2 I_n$:

$$\text{Var}(\widehat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top (\sigma^2 I_n) X (X^\top X)^{-1} = \sigma^2 (X^\top X)^{-1} X^\top X (X^\top X)^{-1}$$

$$\boxed{= \sigma^2 (X^\top X)^{-1}}$$

With $\boxed{\widehat{\sigma}^2 = \dfrac{\sum_i \widehat{\varepsilon}_i^2}{n-k}}$, the estimated covariance matrix is $\boxed{\widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) = \widehat{\sigma}^2 (X^\top X)^{-1}}$.

From the Covariance Matrix to t-Statistics

Diagonal entries give coefficient variances: $\boxed{\widehat{\mathrm{Var}}(\hat{\beta}_j) = [\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}})]_{jj}}$, hence

$$\boxed{SE(\hat{\beta}_j) = \sqrt{[\widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}})]_{jj}}}, \qquad \boxed{t_j = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}} \xrightarrow{H_0:\beta_j=0} \boxed{t_j = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}}$$

$$\boxed{\text{error covariance assumption} \rightarrow \widehat{\mathrm{Var}}(\hat{\boldsymbol{\beta}}) \rightarrow SE(\hat{\beta}_j) \rightarrow t_j \rightarrow p\text{-value, CI}}$$

What If the Error Covariance Is Misspecified?

If instead $\boxed{\mathrm{Var}(\varepsilon \mid X) = \Omega \neq \sigma^2 I_n}$, the *correct* covariance of $\widehat{\boldsymbol{\beta}}$ is

$$\boxed{\mathrm{Var}(\widehat{\boldsymbol{\beta}} \mid X) = (X^\top X)^{-1} X^\top \Omega X (X^\top X)^{-1}}$$

and the classical formula $\widehat{\sigma}^2 (X^\top X)^{-1}$ is generally **no longer valid**.

Heteroskedasticity

$\mathrm{Var}(\varepsilon_i \mid X) = \sigma_i^2$ varies by $i \Rightarrow \Omega = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2) \neq \sigma^2 I_n$.

Serial correlation

$\mathrm{Cov}(\varepsilon_t, \varepsilon_{t-h} \mid X) \neq 0 \Rightarrow \Omega$ has non-zero off-diagonal entries.

Coefficient Estimation vs. Statistical Inference

Under exogeneity $\boxed{\mathbb{E}[\varepsilon \mid X] = 0}$:

$$\mathbb{E}[\widehat{\boldsymbol{\beta}} \mid X] = \boldsymbol{\beta} + (X^\top X)^{-1} X^\top \mathbb{E}[\varepsilon \mid X] = \boxed{\boldsymbol{\beta}}$$

so heteroskedasticity/serial correlation do **not** by themselves bias $\widehat{\boldsymbol{\beta}}$. But the covariance estimator can still be wrong:

$$\boxed{\text{wrong } \Omega \Rightarrow \text{wrong } \widehat{\text{Var}}(\widehat{\boldsymbol{\beta}}) \Rightarrow \text{wrong } SE \Rightarrow \text{wrong } t_j \Rightarrow \text{wrong } p\text{-value/CI}}$$

Numerical Illustration: Same $\widehat{\beta}$, Different Inference

Suppose $\widehat{\beta}_j = 0.10$. With the classical $SE_{\text{classical}} = 0.04$: $t = 0.10/0.04 = 2.5$ (looks significant). With a robust $SE_{\text{robust}} = 0.07$:

$$t_{\text{robust}} = \frac{0.10}{0.07} \approx 1.43$$

The point estimate is unchanged, $\boxed{\widehat{\beta}_j = 0.10}$, but the conclusion about significance changes – because our estimate of *uncertainty* around it changed, not the estimate itself.

Heteroskedasticity-Robust and HAC Standard Errors

If heteroskedasticity is suspected, use a heteroskedasticity-consistent covariance estimator (e.g. HC3): coefficients are unchanged,

$$\boxed{\widehat{\boldsymbol{\beta}}_{\mathrm{HC3}} = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}} \qquad \text{but} \qquad \boxed{\widehat{\mathrm{Var}}_{\mathrm{HC3}}(\widehat{\boldsymbol{\beta}}) \neq \widehat{\sigma}^2 (X^\top X)^{-1}}$$

When residuals exhibit **both** heteroskedasticity **and** serial correlation (common in time series), a **HAC / Newey–West** estimator makes the covariance robust to both, up to a chosen lag structure – again leaving $\widehat{\boldsymbol{\beta}}$ unchanged while correcting SE's, t's, and p-values.

OLS, WLS, GLS: A Hierarchy of Estimators

OLS assumes $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$ and minimizes

$$\boxed{\widehat{\beta}_{\mathrm{OLS}} = \arg\min_{\beta} \sum_i (y_i - \mathrm{x}_i^\top \beta)^2}$$

WLS (heteroskedasticity) down-weights noisy observations, $w_i \propto 1/\sigma_i^2$:

$$\boxed{\widehat{\beta}_{\mathrm{WLS}} = \arg\min_{\beta} \sum_i w_i (y_i - \mathrm{x}_i^\top \beta)^2}$$

GLS: The General Case

GLS allows a fully general error covariance $\boxed{\text{Var}(\varepsilon \mid X) = \Omega}$ (unequal variances *and* correlations):

$$\boxed{\widehat{\beta}_{\text{GLS}} = (X^\top \Omega^{-1} X)^{-1} X^\top \Omega^{-1} \text{y}}$$

WLS is the special case $\boxed{\Omega = \text{diag}(\sigma_1^2, \ldots, \sigma_n^2)}$ (diagonal, no correlations). Hence the hierarchy

$$\boxed{\text{OLS} \subset \text{WLS} \subset \text{GLS}}$$

– OLS: equal variances; WLS: observation-specific variances; GLS: fully general covariance structure.